Health & Social Care in Hansard: a topic-pilot brief

The pipeline, the pilot test, and the substantive checks we now need

July 2026

1. The goal

We want to measure what Parliament has said about health and social care (H&SC) across the full Hansard record, 1803 to the present. For every speech we want to know two things: whether it substantively engages H&SC, and if so which aspects of it the speaker raises. With that in place we can trace how the volume and the mix of H&SC debate shift over two centuries, by era and by topic. The obstacle is scale. The record runs to millions of speeches, far more than anyone can read, so the measurement has to be done by machine and then trusted. This note describes the pipeline we have built to do that, the pilot we ran to test it, and the substantive questions we now need domain input on before scaling up.

2. The main pipeline, step by step

The production pipeline has four stages. Each exists for a reason worth stating, because the reasons are also where the method could go wrong.

1. Narrow the corpus to plausible candidates. The large majority of Hansard never touches H&SC, so running a language model on every speech would be wasteful. In production this stage will be a semantic search over sentence embeddings, retrieving speeches close in meaning to H&SC whatever their exact wording, and that retrieval step will itself need validation. For now we use a broad keyword net (NHS and public-health terms for the modern era; Poor Law, workhouse, sanitary-reform and epidemic-disease terms for the nineteenth century) as a deliberate placeholder: it is crude, but it lets us build and test the rest of the pipeline without waiting on the retrieval work. Either way this stage decides only what gets looked at, never the answer, and it is deliberately over-inclusive. It leaves a few hundred thousand candidate speeches rather than millions.
2. Read each candidate with a language model. For every surviving speech the model answers three questions: does this speech substantively discuss H&SC (yes or no); if so, which sub-topics does it raise; and a supporting verbatim quote. The judgement needs to weigh meaning in context, which a keyword cannot do. The sub-topics are free text, written by the model rather than chosen from a fixed list, because the aim is to discover what is actually discussed rather than to score speeches against categories we set in advance.
3. Aggregate the free-text sub-topics into a taxonomy. The models describe the same idea in many wordings, so to count topics we group same-meaning phrases automatically and label each group. Section 5 explains how, and why this step is more delicate than it looks.
4. Turn the labelled topics into measures over time. The payoff is prevalence: how much of H&SC debate each topic occupies, and how that changes across the 220 years.

Stage 2 sends the model a short instruction, an output-format request, and the speech. The core elements are reproduced verbatim below; the pilot varies their surface form (Section 3) but not their meaning. The topic definition is carried inside the instruction, which makes it easy to overlook as a piece of wording. It is in fact the most consequential setting in the pipeline, and Section 5 shows what happens when it changes.

Instruction (one of two paraphrases):

> Determine whether the parliamentary speech below substantively discusses health and social care — that is, UK health and social care policy: the NHS, public and mental health, adult social care, care homes, and the people who provide or rely on that care — rather than merely mentioning it in passing. If it does, identify the specific sub-topics of health and social care that it discusses. Describe each sub-topic in your own words as a short phrase (a few words). Give at most 5 sub-topics. Also provide the single most relevant verbatim quotation from the speech as supporting evidence.

Output format (the structured condition):

> Respond with a single JSON object and nothing else, with exactly these keys: "mentions_topic" (true or false), "subthemes" (an array of short free-text phrases, empty if mentions_topic is false), and "evidence_quote" (a string, empty if mentions_topic is false).

The other output condition gives no format instruction at all, leaving the model to answer in free text. When present, an optional system message adds a role: "You are an expert analyst of health and social care policy in the United Kingdom, with a deep understanding of parliamentary debate."

3. The pilot: a test before scaling

Running stages 1 to 4 on the whole corpus is expensive and hard to undo, so we first tested the approach on a sample of 270 speeches, stratified across eras and speech lengths. The pilot adds one thing the production run will not use: a robustness harness. Instead of reading each speech once, we read it many ways and check that the answer is driven by the speech rather than by incidental choices of model or wording. If the answers move a lot under variation, the production numbers are not yet trustworthy.

Concretely, each speech was read by four models (three production candidates plus one larger reference model) under eight wordings of the same request, varying the role framing, the paraphrase, and the output format. That is 32 independent reads per speech. Two results frame everything that follows:

- On the yes/no presence question the reads agree with each other about 82% of the time, which is solid but short of unanimous.
- The share of speeches counted as being about H&SC ranges from 34% to 81% depending on prompt and model, averaging around 51%. Much of that swing comes from the output format: asking for free-text output puts presence at about 55%, while asking for structured output puts it at about 43%. The headline rate is therefore sensitive to prompt design, which is one reason the substantive checks below matter.

4. What the models found: the topic map

Across the 270 speeches the models emitted 8,692 distinct sub-topic phrases. Grouped by meaning, the most common topics are shown below. Both modern NHS-era themes and nineteenth-century themes such as Poor Law, sanitation and housing appear. This coverage should be read with the sampling in mind: the keyword net was deliberately built to include both modern and historical H&SC vocabulary, so the presence of both is partly a consequence of how the sample was seeded rather than independent evidence about the balance of the corpus. What the map does show is what the models surface as sub-topics once a speech is placed in front of them, and that is what the rest of this note examines.

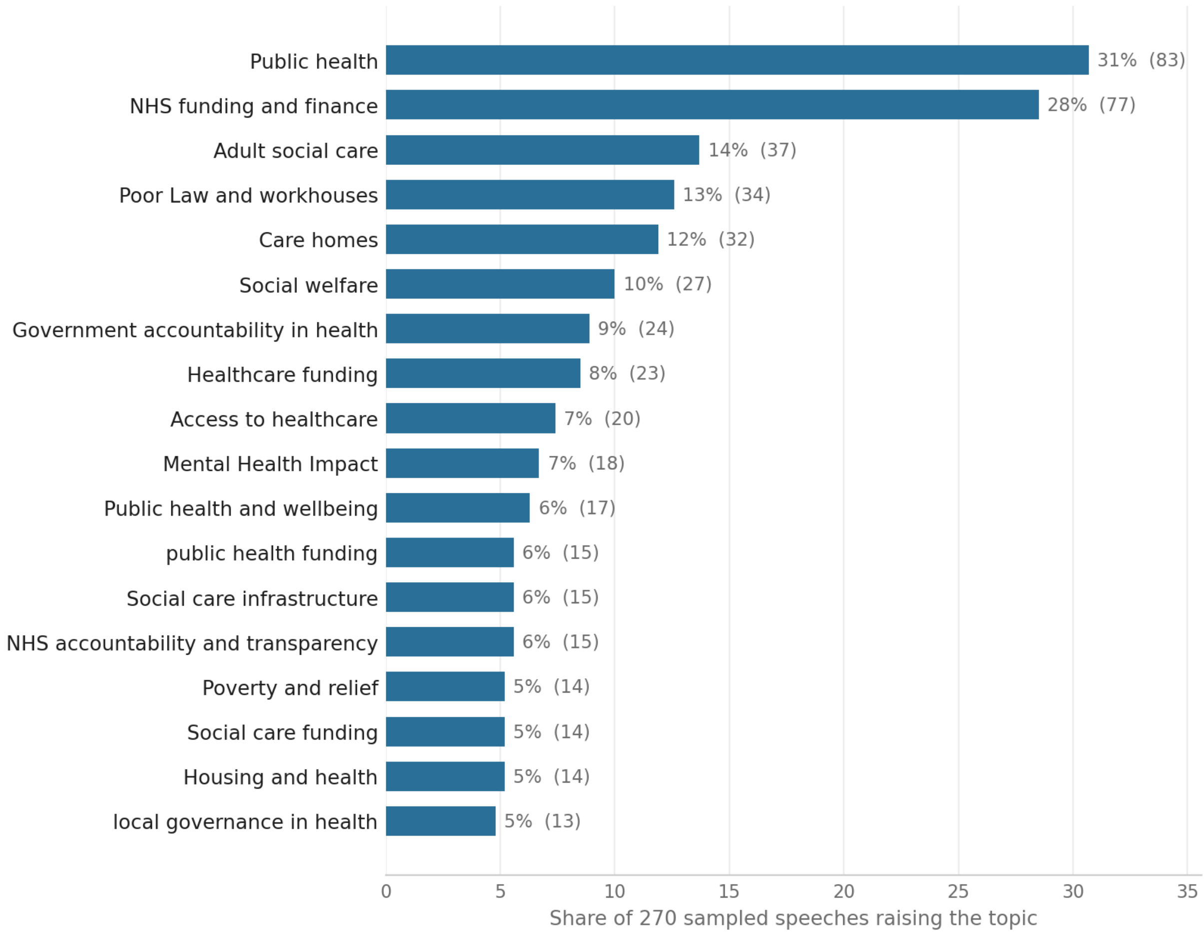

Figure 1: Most common H&SC topics across the 270-speech pilot (share of speeches raising each; count in brackets). Labels are the clusters' automatically chosen medoid phrases, not hand-edited.

Each row below is a machine-discovered topic, that is, a cluster of same-meaning phrases, ranked by how many of the 270 speeches raise it. The "Phr." column counts how many distinct wordings the models used for that one idea. The label in each row is the cluster's medoid: the single emitted phrase closest to the cluster's centre in meaning-space, chosen automatically rather than written by us. Nothing here is hand-edited, which is why some labels read awkwardly or are oddly cased (for example "Mental Health Impact" or "local governance in health"). Naming the topics well is one of the domain judgements we are asking for.

#	Merged topic	Speeches	Phr.	Example sub-themes the models wrote
1	Public health	83 (31%)	99	Public health policy; Public health infrastructure; Public health response
2	NHS funding and finance	77 (28%)	365	NHS funding; NHS; National Health Service
3	Adult social care	37 (14%)	39	Adult social care provision; Adult social care governance; Adult social care regulation
4	Poor Law and workhouses	34 (13%)	187	Poor Law administration; Poor Law; Workhouse conditions
5	Care homes	32 (12%)	38	Institutional care; Institutional care conditions; Care homes for old people
6	Social welfare	27 (10%)	33	Social welfare provision; Social welfare policy; Social welfare funding

#	Merged topic	Speeches	Phr.	Example sub-themes the models wrote
7	Government accountability in health	24 (9%)	61	Public health governance; Government accountability in health policy; Government transparency in health
8	Healthcare funding	23 (8%)	31	healthcare financing; Health service funding; Hospital Funding
9	Access to healthcare	20 (7%)	38	Access to care; Healthcare access; Access to treatment
10	Mental Health Impact	18 (7%)	15	Mental health; Mental health impacts; Mental health effects
11	Public health and wellbeing	17 (6%)	8	Public or mental health; Public or mental health policy; Public health (mental wellbeing)
12	public health funding	15 (6%)	30	Public health financing; Public health finance; Public health funding and expenditure
13	Social care infrastructure	15 (6%)	16	Social care provision; Social care policy; Social care provisions
14	NHS accountability and transparency	15 (6%)	41	Health service accountability; NHS accountability; NHS accountability and governance
15	Poverty and relief	14 (5%)	20	Poverty relief; Poverty relief management; Poverty and destitution
16	Social care funding	14 (5%)	49	Adult social care funding; Public funding of care; Funding for social care services
17	Housing and health	14 (5%)	47	Housing conditions affecting health; Housing and public health; Public health and housing conditions
18	local governance in health	13 (5%)	48	local health governance; local health planning; Local governance of health
19	Health workforce	13 (5%)	25	Healthcare workforce; Health workforce planning; Medical workforce planning
20	Mental health services and community care	13 (5%)	40	Mental health services; Mental health support; Mental health provision
21	Affordable housing	12 (4%)	29	Housing; Housing for the working classes; Housing for the poor
22	Public funding allocation	12 (4%)	21	Government funding; Public funding; Public Expenditure
23	Sanitary conditions	12 (4%)	31	public health and sanitation; Sanitation and hygiene; Sanitation and disease prevention
24	Local authority powers	12 (4%)	22	Local authority responsibility; Local authority role; Local authority duties
25	Care home regulation	11 (4%)	29	Social care regulation; Care home regulation and administration; Care homes regulation
26	NHS employment	11 (4%)	68	NHS workforce; NHS recruitment; NHS workforce planning
27	Health legislation	11 (4%)	10	Public health legislation; Healthcare legislation; Health service legislation
28	Infectious disease control	11 (4%)	20	Infectious disease; Infection control; Infectious disease prevention
29	Healthcare administration	11 (4%)	11	Health administration; Health service administration; Healthcare management
30	Healthcare access inequality	10 (4%)	31	Healthcare inequality; Health inequalities; Inequality in healthcare access
31	Local administration and governance of care	10 (4%)	16	Governance of care institutions; Governance of social care services; Social care governance
32	Housing conditions	10 (4%)	19	Living conditions; Housing and living conditions; Housing standards
33	Healthcare resource allocation	10 (4%)	19	resource allocation in health; Resource allocation for healthcare; Health Resource Allocation
34	Health policy	10 (4%)	11	Healthcare policy; health policy consultation; Health policy measures
35	Government accountability	10 (4%)	7	Government Responsibility; Political accountability; Democratic accountability
36	Public health campaigns	9 (3%)	16	public health initiatives; Public health awareness campaigns; Public health communication
37	NHS response to infection	9 (3%)	19	NHS response; NHS response to outbreaks; NHS response to pandemic threat
38	Public health and prevention	9 (3%)	25	Public health prevention; Preventative healthcare; preventive healthcare

#	Merged topic	Speeches	Phr.	Example sub-themes the models wrote
39	Health care charges	9 (3%)	30	Healthcare charges; Healthcare costs; Health service costs
40	Disease control	9 (3%)	8	Disease prevention; Disease control measures

A long tail of some 700 smaller topics sits below this list and grows increasingly specific. It is available on request.

5. Where the sub-topics come from, and why aggregation is delicate

The topic map is not a neutral readout of what Parliament said. It is shaped at several points by choices we made, each of which the pilot suggests we should revisit with domain input.

The definition sets the era profile. Everything else in this note is measured against one definition of H&SC, the one quoted in Section 2, which names the NHS, adult social care and care homes. Those are institutions that did not exist for the first century and a half of the record, so the wording risks reading the nineteenth century as quiet about health simply because it lacked the vocabulary. We tested this by re-running the same speeches, models and formats against an era-neutral rewrite that describes the function rather than the institution: the health of the population and the care of people who are sick, injured, disabled, elderly or destitute, however that care is provided and paid for. The effect is large, and concentrated exactly where the concern predicted. Under the current definition the share of speeches judged H&SC climbs from 33% before 1900 to 62% after 1948; under the era-neutral wording it runs from 54% to 67%, and the rise across the record falls from 29 to 13 percentage points (Figure 2). The two definitions agree on 82% of individual reads, and the disagreements run almost entirely one way: the era-neutral wording adds speeches, in 23% of pre-1900 reads but only 7% of post-1948 ones. On this evidence a good part of the apparent growth in health debate after the NHS is an artefact of how we asked the question.

This does not establish that the era-neutral wording is the right one. Reading the speeches it newly admits shows both kinds of case. Some are plain misses by the current definition: an 1868 exchange in which the Poor Law Board sets standards for the diet, lodging and work of vagrants is unmistakably the Victorian care system, and the current wording scored it as H&SC in only one read out of eight. Others look like over-reach: an 1856 debate about which body should hold powers under an agricultural statistics bill was counted as H&SC by seven of eight reads because the Poor Law Board happens to be named in it, which is precisely the passing mention the instruction rules out. That second case is worth stating plainly, because it is the same failure as before rather than a new one. The old wording triggered on the token NHS; the new one triggers on tokens like Poor Law Board. Changing the definition moved which century the anchoring flatters, but did not stop the models keying on institution names instead of on whether care is the subject of the speech. Choosing between the two wordings, and repairing whichever we keep so that neither trap fires, is a domain judgement and not something a further run can settle.

The prompt's output format changes the result. Each read was asked either for free-text output or for structured output, and this matters more than expected. The number of sub-topics per speech is unaffected (about 3.8 either way, often hitting the cap of five), so the format does not squeeze the count. What it changes is the wording and the presence rate. Free-text output produced 5,903 distinct phrasings; structured output produced only 3,639, roughly 40% fewer, because structured output regularises how the model names the same idea while free text lets it drift. Free-text output also raised the presence rate (55% versus 43%) and failed to parse cleanly about 5% of the time, which is the source of the noise described below. So the raw phrase counts in the table, and how badly topics fragment, are partly a property of prompt design rather than

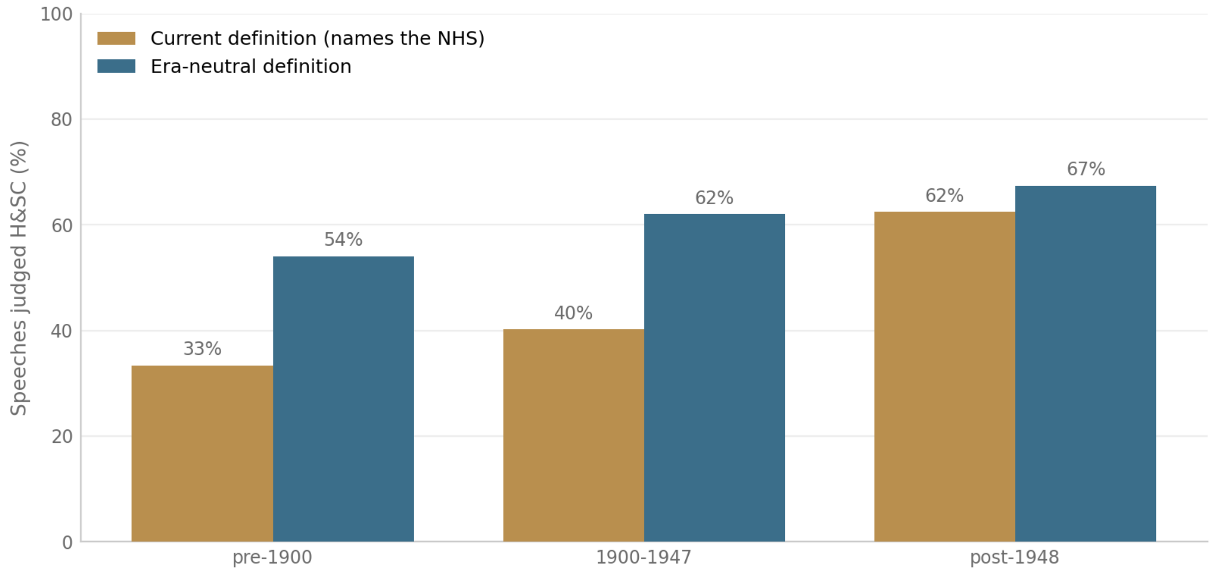

Figure 2: Share of speeches judged to be about H&SC, by era, under the two definitions. Neither is known to be correct; the point is that the choice of wording, not the record, sets much of the slope.

of the speeches. In production we will fix a single output format, and that is itself a measurement decision, not just an engineering one.

The five-topic cap inflates the count. The instruction asks for at most five sub-topics, and we have now tested removing it: the same speeches, models and formats, run once with the "at most five" sentence and once without. The cap turns out to work as an anchor, not just a ceiling. With the cap, half of all positive reads return exactly five sub-topics and the median is five. Without it, that spike disappears (only 9% land on five), the distribution settles onto a smooth curve centred on two to three, and the median drops to three (Figure 3). About 14% of speeches genuinely warrant more than five sub-topics when allowed, up to eighteen, but those are the minority; the dominant effect is the cap pulling the many lower-content speeches up towards five. Presence is unaffected (48% either way), so this is purely about how many sub-topics get counted, not whether a speech is judged to be about H&SC. The practical implication is that any count-based measure built on the capped runs is biased upward, so production should drop the cap or set it far higher.

Aggregation depends on one threshold. Same-meaning phrases are merged by converting each phrase into a numeric representation of its meaning (an embedding) and clustering the ones that sit close together, with each cluster labelled by its most central phrase (its medoid). A single setting controls how readily phrases merge. Set it loose and unrelated ideas get lumped together; set it tight and one idea fragments into several topics. We currently keep it tight to avoid false merges, and the cost of that is visible in the table: Public health (row 1) sits apart from Public health and wellbeing (11), public health funding (12), Public health campaigns (36) and Public health and prevention (38), and the same fragmentation runs through the funding rows (2, 8, 12, 16, 22) and the governance rows (7, 14, 24, 31, 35). Where these should be merged, and how broad the final families should be, is a domain judgement rather than a distance measurement.

A small share of entries are formatting noise, not content. From the free-text parse failures above, a model's answer sometimes leaked into the topic field. These are a known and fixable side effect, not real findings, but they are worth seeing:

- Does the speech substantively discuss health and social care? No

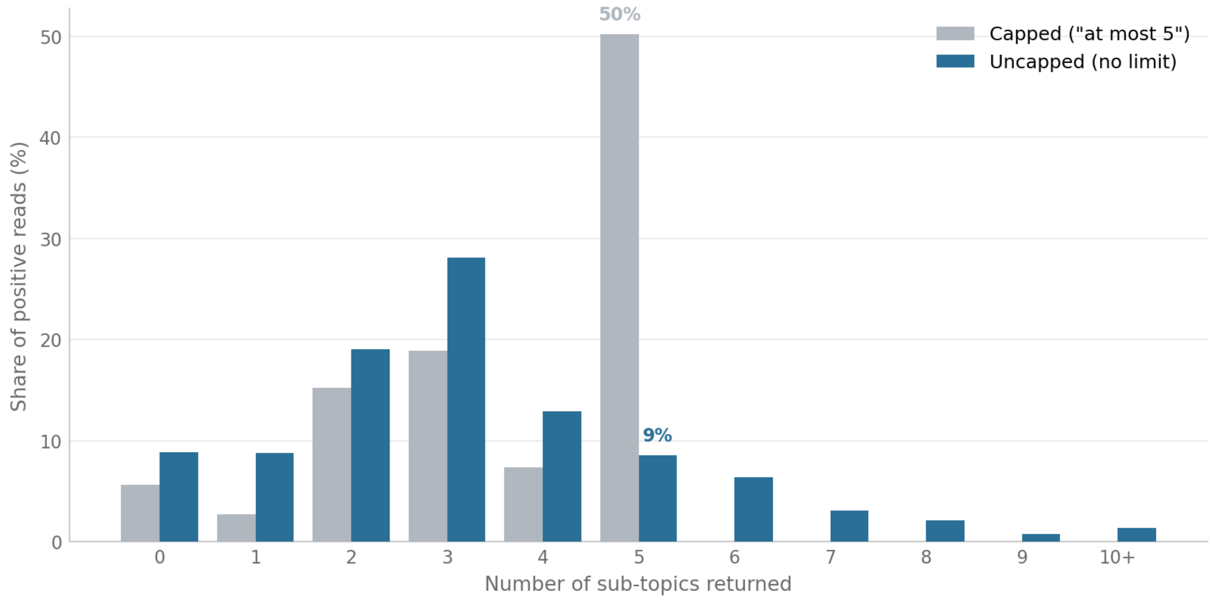

Figure 3: Sub-topics returned per positive read, capped versus uncapped, on the same speeches. The cap concentrates half of all reads at exactly five; without it the count follows a smooth low distribution.

- Sub-topics of health and social care discussed: None
- Most relevant verbatim quotation (for context): …

Taken together, the prevalence numbers depend jointly on the definition, the output format, the five-topic cap, and the clustering threshold. We can now put rough sizes on the first two: the definition moves the era profile by around 16 percentage points of slope, and the output format moves the headline rate by about 12 points. None of the four is settled by the machine, which is precisely why the next step is substantive rather than technical.

6. What we need next

Two pieces of work now need domain input, and they are the reason for this brief.

The first is a substantive review of the current map:

1. Scope. Section 5 shows the definition is the most consequential setting we have found, and the pilot cannot resolve it: the two wordings we tested fail in opposite directions. Which is closer to how the field bounds H&SC, and how would you word it so that neither the NHS nor the Poor Law Board pulls a speech in on the strength of being mentioned?
2. Completeness. Is any important H&SC theme missing that Parliament would have debated across this period?
3. Correctness. Is any listed topic wrong, or wrongly counted as H&SC?
4. Grouping and grain. How far should the over-split topics in Section 5 be merged, and at what level of detail should the final set sit?

The second is designing how the production pipeline will be evaluated, which the paper will stand or fall on. The pilot only shows that the models agree with each other, which is consistency, not that they agree with the field, which is validity. To close that gap we need to build, together:

- A gold standard: a set of speeches hand-labelled by domain experts for presence and for sub-topics. This is the reference the whole system is measured against, and the definition experiment has already produced an efficient starting set. On 159 of the 270 pilot speeches the two definitions reach different verdicts, 92 of them by a wide margin, and those are the speeches where the boundary is actually being decided rather than being obvious. They are laid out in `definition_review.html`, an offline page that shows each speech next to what the models said under both definitions and records a yes, borderline or no judgement with a note, then exports the lot as a spreadsheet. Working through the contested set is worth considerably more per speech than labelling a random sample, in which most cases are easy and carry little information.

- An evaluation protocol we can re-run unchanged whenever a model, prompt, or threshold changes: presence scored against the gold labels, and sub-topics scored on whether the model's topics match the experts' in meaning.
- A decision, informed by that protocol, on the production settings the pilot has shown to matter: the definition, the output format, the sub-topic cap (Section 5 shows the current five is biasing counts), and the clustering grain.

Any form works for a first pass: notes in the margin, a shared document, or a call.

Appendix: reference

A. Filter keywords. The placeholder net from Section 2 matches the terms below, as word stems, in three groups. A speech enters the sample if it matches any of them; the terms decide only what is looked at, never the answer.

- Modern (NHS and welfare state): social care, care home, domiciliary care, residential care, adult social care, care worker, carer, NHS, national health service, public health, mental health, health care, healthcare.
- Pre-NHS (Poor Law and sanitary reform): poor law, workhouse, pauper, board of guardians, relieving officer, district nursing, lunatic asylum, feeble-minded, mental deficiency, friendly society, infirmary, almshouse, sanitary, board of health, fever hospital, vaccination, smallpox, cholera, tuberculosis, medical officer of health, dispensary.
- Epidemic and pandemic: covid, coronavirus, pandemic, long covid, lockdown, furlough, epidemic, influenza, spanish flu, quarantine, outbreak, typhoid, typhus, diphtheria, scarlet fever, plague.

B. Prompt, second paraphrase. Section 2 gives the first wording of the instruction. The pilot also uses this second, meaning-preserving paraphrase:

> Read the parliamentary speech below. Decide if health and social care is a substantive subject of the speech and not just a passing reference. Treat health and social care as: UK health and social care policy: the NHS, public and mental health, adult social care, care homes, and the people who provide or rely on that care. When it is a substantive subject, list the particular aspects of health and social care it covers, each as a brief free-text label of a few words. Provide no more than 5 such labels. Include a short supporting quotation taken word-for-word from the speech.

Role and output format vary as in Section 2 (role present or absent, output structured or free text), giving eight prompt combinations in all.

C. Definition wordings tested. Both are dropped into the same slot in the instruction, after "substantively discusses health and social care, that is,". Everything else in the prompt is identical, so the contrast between them is the definition and nothing else.

> Current: UK health and social care policy: the NHS, public and mental health, adult social care, care homes, and the people who provide or rely on that care.

> Era-neutral: the health of the population and the care of people who are sick, injured, disabled, elderly or destitute, however that care is provided and paid for, whether by

the state, local authorities, hospitals, charities, religious bodies or families.

Two further wordings, one narrower (clinical services only) and one broader (including the social determinants of health such as housing, sanitation and nutrition), are prepared but not yet run. They are intended to bracket the construct so the headline prevalence can be reported as a range rather than a single number, and are worth running once the definition question above is settled.

D. Extended topic list, ranks 41 to 100. This continues the table in Section 4. Of the roughly 700 machine-discovered topics, 141 are raised by four or more speeches.

#	Merged topic	Speeches	Phr.
41	Funding	8 (3%)	8
42	Sewage and sanitation	8 (3%)	31
43	Public trust in the NHS	8 (3%)	13
44	Carer support and well-being	8 (3%)	40
45	Protection of vulnerable populations	8 (3%)	8
46	Regulatory oversight	8 (3%)	13
47	Children's homes and residential care	8 (3%)	45
48	Ministerial oversight in health	7 (3%)	25
49	Health Service	7 (3%)	9
50	Integration of health and social care	7 (3%)	31
51	Health and social care policy coordination	7 (3%)	10
52	Immunisation policy	7 (3%)	37
53	Healthcare Regulation	7 (3%)	19
54	Public financial accountability	7 (3%)	8
55	Patient safety in healthcare	7 (3%)	18
56	Social care and housing for vulnerable groups	7 (3%)	8
57	Workhouse medical care	7 (3%)	18
58	Healthcare reform	7 (3%)	11
59	Hospital design	6 (2%)	22
60	Tuberculosis treatment and control	6 (2%)	41
61	Government oversight in health	6 (2%)	10
62	Local Authority Care	6 (2%)	11
63	Impact of economic policy on health care	6 (2%)	15
64	NHS estate	6 (2%)	50
65	Post-war social needs	6 (2%)	23
66	Old age pensions	6 (2%)	30
67	Government oversight	6 (2%)	6
68	Parliamentary Scrutiny	6 (2%)	8
69	Public participation in healthcare decision-making	6 (2%)	32
70	Public relief systems	6 (2%)	6
71	Healthcare professional support	6 (2%)	11
72	government healthcare planning	6 (2%)	12
73	Public health administration	6 (2%)	5
74	Global public health	6 (2%)	14
75	Hospital care	6 (2%)	5
76	Social security funding	5 (2%)	15
77	Social services management	5 (2%)	22
78	medical care for the poor	5 (2%)	21
79	Patient care	5 (2%)	7
80	Community wellbeing	5 (2%)	5
81	Social care legislation and reform	5 (2%)	14
82	Health care prioritisation	5 (2%)	15
83	Historical health policy	5 (2%)	13
84	Free healthcare at the point of use	5 (2%)	12
85	NHS information provision	5 (2%)	26
86	Parliamentary timing of health legislation	5 (2%)	16
87	Poverty and health	5 (2%)	6
88	Public health and food poverty	5 (2%)	10
89	Public health response to epidemic	5 (2%)	29
90	Armed forces healthcare	5 (2%)	7
91	Support for vulnerable individuals	5 (2%)	4

#	Merged topic	Speeches	Phr.
92	Local authority health responsibilities	5 (2%)	12
93	NHS Commissioning	5 (2%)	20
94	Overcrowding and housing	5 (2%)	16
95	Decision-making processes in care and legal proceedings	5 (2%)	7
96	Local authority housing provision	5 (2%)	14
97	public health policy implementation	5 (2%)	6
98	Local taxation	5 (2%)	6
99	Older people's care	5 (2%)	30
100	Disease outbreaks	5 (2%)	6

E. Most common sub-topic phrases, verbatim, before grouping. These show the raw vocabulary and how much case and wording variation there is.

Sub-theme phrase (verbatim)	Speeches	Total emissions
Public health	45	130
public health	31	65
Public Health	26	84
Adult social care	25	40
NHS funding	23	128
Public health policy	22	37
Care homes	18	30
National Health Service	15	76
NHS	15	28
Social welfare	14	21
Poor Law administration	13	41
Poor Law	12	22
Public health infrastructure	12	25
NHS governance	10	30
Mental health	9	15
Workhouse conditions	7	40
Disease prevention	7	26
Housing conditions	7	27
Mental health services	7	29
Patient safety	7	21

F. Pilot facts. 270 speeches, read by four models under eight prompt wordings each (32 reads per speech at temperature zero), producing 8,692 distinct sub-topic phrases and 15,634 total phrase emissions. That core grid is accompanied by two controlled experiments of 2,160 reads each, matched to it on speech, model and format: the uncapped arm behind Section 5's cap finding, and the era-neutral arm behind the definition finding. The sample spans 1803 to the present. Sub-topics are generated by the models rather than chosen from a list. All figures come from the current pilot run.